

Appendix to

Search Frictions and the Business Cycle in a Small Open Economy DSGE Model

Juan Guerra-Salas[†] Markus Kirchner[‡] Rodrigo Tranamil[§]

June 2020

Note: This appendix is not a self-contained document.

Contents

A	Equilibrium Conditions of the Search Model	1
B	Steady State of the Search Model	5
C	Equilibrium Conditions of the Calvo Wages Model	10
D	Steady State of the Calvo Wages Model	11
E	Detailed Results and Robustness Checks	13

[†]Central Bank of Chile, Monetary Policy Division. E-mail address: jguerra@bcentral.cl.

[‡]Central Bank of Chile, Financial Policy Division. E-mail address: mkirchner@bcentral.cl.

[§]Central Bank of Chile and Universitat Pompeu Fabra. E-mail address: rtranamil@bcentral.cl.

A Equilibrium Conditions of the Search Model

The variables in uppercase that are not prices contain a unit root in equilibrium due to the presence of the non-stationary productivity shock. To have a stationary version of the model, we divided those variables in t by A_{t-1} . For the case of the Lagrange multiplier Λ_t , we multiply it by A_{t-1}^σ because it decreases along the balanced growth path.

The rational expectations equilibrium of the stationary version of the model is the set of sequences (60 variables)

$$\left\{ \begin{array}{l} \lambda_t, q_t, c_t, i_t, k_t, \Theta_t, \tilde{\chi}_t, u_t, m_t, n_t, \rho_t, s_t, e_t, \bar{\mathcal{E}}_t, \mathcal{E}_t^*, \mathcal{U}_t, \Delta_t^1, \Delta_t^2, w_t^*, \epsilon_t, \mu_t, \mathcal{F}_t^*, \mathcal{W}_t^* \\ \bar{W}_t, w_t, \bar{w}_t, \tilde{w}_t, h_t, y_t^C, x_t^H, x_t^F, y_t^H, y_t^F, p_t^w, \tilde{p}_t^H, \tilde{p}_t^F, f_t^H, f_t^F, mc_t^H, mc_t^F, r_t^K \\ v_t, \rho_t^n, \bar{f}_t, \psi_t, y_t^m, \pi_t, R_t, \xi_t, rer_t, \pi_t^s, x_t^{H*}, p_t^H, p_t^F, \Delta_t^H, \Delta_t^F, b_t^*, tb_t, y_t, p_t^Y \end{array} \right\}_{t=0}^\infty$$

such that for given initial values and exogenous sequences

$$\{\kappa_t, \varrho_t, \varpi_t, \rho_t^x, z_t, a_t, y_t^{Co}, g_t, R_t^*, \pi_t^*, \zeta_t, y_t^*, p_t^{Co*}\}_{t=0}^\infty$$

and assuming $F_t \sim \log N(0, \sigma_w^2)$, the following conditions are satisfied:

Household (5 equations)

$$\lambda_t = \left(c_t - \zeta \frac{c_{t-1}}{a_{t-1}} \right)^{-\sigma} \quad (1)$$

$$\begin{aligned} \frac{1}{q_t} = & \left[1 - \frac{\gamma}{2} \left(\frac{i_t}{i_{t-1}} a_{t-1} - \bar{a} \right)^2 - \gamma \left(\frac{i_t}{i_{t-1}} a_{t-1} - \bar{a} \right) \frac{i_t}{i_{t-1}} a_{t-1} \right] \varpi_t \\ & + \frac{\beta}{a_t^\sigma} \gamma E_t \left\{ \frac{\varrho_{t+1}}{\varrho_t} \frac{\lambda_{t+1}}{\lambda_t} \frac{q_{t+1}}{q_t} \left(\frac{i_{t+1}}{i_t} a_t - \bar{a} \right) \left(\frac{i_{t+1}}{i_t} a_t \right)^2 \varpi_{t+1} \right\} \end{aligned} \quad (2)$$

$$q_t = \frac{\beta}{a_t^\sigma} E_t \left\{ \frac{\varrho_{t+1}}{\varrho_t} \frac{\lambda_{t+1}}{\lambda_t} [r_{t+1}^K + q_{t+1}(1 - \delta)] \right\} \quad (3)$$

$$\lambda_t = \frac{\beta}{a_t^\sigma} R_t E_t \left\{ \frac{\varrho_{t+1}}{\varrho_t} \frac{\lambda_{t+1}}{\pi_{t+1}} \right\} \quad (4)$$

$$k_t = (1 - \delta) \frac{k_{t-1}}{a_{t-1}} + \left(1 - \frac{\gamma}{2} \left(\frac{i_t}{i_{t-1}} a_{t-1} - \bar{a} \right)^2 \right) \varpi_t i_t \quad (5)$$

Preference shifter (2 equations)

$$\Theta_t = \tilde{\chi}_t \left(c_t - \zeta \frac{c_{t-1}}{a_{t-1}} \right)^{-\sigma} \quad (6)$$

$$\tilde{\chi}_t = \tilde{\chi}_{t-1}^{1-v} \left(c_t - \zeta \frac{c_{t-1}}{a_{t-1}} \right)^{\sigma v} \quad (7)$$

Labor market variables (6 equations)

$$m_t = \bar{m} v_t^{1-\mu} u_t^\mu \quad (8)$$

$$n_t = (1 - \rho_t) (n_{t-1} + m_{t-1}) \quad (9)$$

$$u_t = 1 - n_t \quad (10)$$

$$s_t = m_t / u_t \quad (11)$$

$$e_t = m_t / v_t \quad (12)$$

$$\rho_t = \rho_t^n + (1 - \rho_t^n) \rho_t^x \quad (13)$$

Average value of employment (5 equations): recall that $\Xi_{t,t+1} = \frac{\beta}{a_t^\sigma} \frac{\varrho_{t+1}}{\varrho_t} \frac{\lambda_{t+1}}{\lambda_t}$ and $\Gamma_t^W = a_t^{\alpha_w} \pi_t^{\vartheta_w} \bar{\pi}^{1-\vartheta_w}$

$$\bar{\mathcal{E}}_t = w_t h_t - \frac{\Theta_t \kappa_t h_t^{1+\phi}}{\lambda_t (1+\phi)} + a_t E_t \Xi_{t,t+1} [(1 - \rho_{t+1}) (\mathcal{E}_{t+1}^* + \theta^w \Delta_{t+1}^1) + \rho_{t+1} \mathcal{U}_{t+1}] \quad (14)$$

$$\mathcal{E}_t^* = w_t^* h_t - \frac{\Theta_t \kappa_t h_t^{1+\phi}}{\lambda_t (1+\phi)} + a_t E_t \Xi_{t,t+1} [(1 - \rho_{t+1}) (\mathcal{E}_{t+1}^* + \theta^w \Delta_{t+1}^2) + \rho_{t+1} \mathcal{U}_{t+1}] \quad (15)$$

$$\mathcal{U}_t = \bar{b} + a_t E_t \Xi_{t,t+1} [s_t (1 - \rho_{t+1}) \bar{\mathcal{E}}_{t+1} + (1 - s_t (1 - \rho_{t+1})) \mathcal{U}_{t+1}] \quad (16)$$

$$\Delta_t^1 = \left(\frac{\Gamma_{t-1}^W}{a_{t-1}} \frac{w_{t-1}}{w_t^*} - 1 \right) w_t^* h_t + a_t \theta^w E_t \Xi_{t,t+1} \Delta_{t+1}^1 \quad (17)$$

$$\Delta_t^2 = \left(\frac{\Gamma_{t-1}^W}{a_{t-1}} \frac{w_{t-1}^*}{w_t^*} - 1 \right) w_t^* h_t + a_t \theta^w E_t \Xi_{t,t+1} \Delta_{t+1}^2 \quad (18)$$

Negotiated wage (6 equations):

$$\epsilon_t = h_t + \theta^w \Gamma_t^W E_t \Xi_{t,t+1} (1 - \rho_{t+1}) \frac{\epsilon_{t+1}}{\pi_{t+1}} \quad (19)$$

$$\mu_t = h_t + \theta^w \Gamma_t^W (1 - e_t) E_t \Xi_{t,t+1} (1 - \rho_{t+1}) \frac{\mu_{t+1}}{\pi_{t+1}} \quad (20)$$

$$\mathcal{F}_t^* = p_t^w (1 - \alpha) \frac{y_t^H}{n_t} - w_t^* h_t - \psi_t + \frac{\bar{\omega}}{e_t} \quad (21)$$

$$\mathcal{W}_t^* = \frac{\varphi}{1 - \varphi} \frac{\epsilon_t}{\mu_t} \mathcal{F}_t^* \quad (22)$$

$$\bar{\mathcal{W}}_t = \bar{\mathcal{E}}_t - \bar{b} - a_t E_t \Xi_{t,t+1} [s_t (1 - \rho_{t+1}) \bar{\mathcal{W}}_{t+1} + \mathcal{U}_{t+1}] \quad (23)$$

$$\mathcal{W}_t^* = w_t^* h_t - \frac{\Theta_t \kappa_t h_t^{1+\phi}}{\lambda_t (1+\phi)} - \bar{b} + a_t E_t \Xi_{t,t+1} (1 - \rho_{t+1}) \left(\theta^w \frac{\varphi}{1 - \varphi} \frac{\epsilon_{t+1}}{\mu_{t+1}} (1 - \Gamma_{t+1}^W) w_{t+1}^* h_{t+1} - s_t \bar{\mathcal{W}}_{t+1} + \mathcal{W}_{t+1}^* \right) \quad (24)$$

Average real wage (3 equations):

$$w_t = (1 - \theta^w) w_t^* + \theta^w \bar{w}_t \quad (25)$$

$$\bar{w}_t \pi_t = \frac{\Gamma_{t-1}^W}{a_{t-1} n_t} (1 - \rho_t) ((1 - s_{t-1}) w_{t-1} n_{t-1} + s_{t-1} \tilde{w}_{t-1}) \quad (26)$$

$$\tilde{w}_t \pi_t = (1 - \theta^w) w_t^* \pi_t + \frac{\theta^w \Gamma_{t-1}^W}{a_{t-1}} \tilde{w}_{t-1} \quad (27)$$

Individual hours (1 equation):

$$h_t = \left(p_t^w (1 - \alpha)^2 \frac{\lambda_t}{\Theta_t \kappa_t} \frac{y_t^H}{n_t} \right)^{\frac{1}{1+\phi}} \quad (28)$$

Final and composite goods (5 equations):

$$y_t^C = \left[(1 - o)^{\frac{1}{\eta}} (x_t^H)^{\frac{\eta-1}{\eta}} + o^{\frac{1}{\eta}} (x_t^F)^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}} \quad (29)$$

$$x_t^H = (1 - o) (p_t^H)^{-\eta} y_t^C \quad (30)$$

$$x_t^F = o (p_t^F)^{-\eta} y_t^C \quad (31)$$

$$y_t^H \Delta_t^H = z_t \left(\frac{k_{t-1}}{a_{t-1}} \right)^\alpha (a_t n_t h_t)^{1-\alpha} \quad (32)$$

$$y_t^F \Delta_t^F = y_t^m \quad (33)$$

Calvo-pricing of retail goods (9 equations): recall that $\Gamma_t^H = \pi_{t-1}^{\vartheta_H} \bar{\pi}^{1-\vartheta_H}$ and $\Gamma_t^F = \pi_{t-1}^{\vartheta_F} \bar{\pi}^{1-\vartheta_F}$

$$mc_t^H = p_t^w / p_t^H \quad (34)$$

$$mc_t^F = rer_t / p_t^F \quad (35)$$

$$f_t^H = (\tilde{p}_t^H)^{-\epsilon_H} mc_t^H y_t^H + a_t \theta_H E_t \Xi_{t,t+1} \left\{ \left(\frac{\Gamma_t^H}{\pi_{t+1}} \right)^{-\epsilon_H} \left(\frac{\tilde{p}_t^H}{\tilde{p}_{t+1}^H} \right)^{-\epsilon_H} \left(\frac{p_t^H}{p_{t+1}^H} \right)^{-1-\epsilon_H} f_{t+1}^H \right\} \quad (36)$$

$$f_t^H = (\tilde{p}_t^H)^{1-\epsilon_H} \left(\frac{\epsilon_H - 1}{\epsilon_H} \right) y_t^H + a_t \theta_H E_t \Xi_{t,t+1} \left\{ \left(\frac{\Gamma_t^H}{\pi_{t+1}} \right)^{1-\epsilon_H} \left(\frac{\tilde{p}_t^H}{\tilde{p}_{t+1}^H} \right)^{1-\epsilon_H} \left(\frac{p_t^H}{p_{t+1}^H} \right)^{-\epsilon_H} f_{t+1}^H \right\} \quad (37)$$

$$f_t^F = (\tilde{p}_t^F)^{-\epsilon_F} mc_t^F y_t^F + a_t \theta_F E_t \Xi_{t,t+1} \left\{ \left(\frac{\Gamma_t^F}{\pi_{t+1}} \right)^{-\epsilon_F} \left(\frac{\tilde{p}_t^F}{\tilde{p}_{t+1}^F} \right)^{-\epsilon_F} \left(\frac{p_t^F}{p_{t+1}^F} \right)^{-1-\epsilon_F} f_{t+1}^F \right\} \quad (38)$$

$$f_t^F = (\tilde{p}_t^F)^{1-\epsilon_F} \left(\frac{\epsilon_F - 1}{\epsilon_F} \right) y_t^F + a_t \theta_F E_t \Xi_{t,t+1} \left\{ \left(\frac{\Gamma_t^F}{\pi_{t+1}} \right)^{1-\epsilon_F} \left(\frac{\tilde{p}_t^F}{p_{t+1}^F} \right)^{1-\epsilon_F} \left(\frac{p_t^F}{p_{t+1}^F} \right)^{-\epsilon_F} f_{t+1}^F \right\} \quad (39)$$

$$1 = (1 - \theta_H) (\tilde{p}_t^H)^{1-\epsilon_H} + \theta_H \left(\frac{p_{t-1}^H}{p_t^H} \frac{\Gamma_t^H}{\pi_t} \right)^{1-\epsilon_H} \quad (40)$$

$$1 = (1 - \theta_F) (\tilde{p}_t^F)^{1-\epsilon_F} + \theta_F \left(\frac{p_{t-1}^F}{p_t^F} \frac{\Gamma_t^F}{\pi_t} \right)^{1-\epsilon_F} \quad (41)$$

$$\frac{\bar{\omega}}{e_t} = a_t E_t \Xi_{t,t+1} (1 - \rho_{t+1}) \left(p_{t+1}^w (1 - \alpha) \frac{y_{t+1}^H}{n_{t+1}} - w_{t+1} h_{t+1} - \psi_{t+1} + \frac{\bar{\omega}}{e_{t+1}} \right) \quad (42)$$

Intermediate goods firms (6 equations):

$$r_t^K = p_t^w \alpha \frac{y_t^H}{k_{t-1}} a_{t-1} \quad (43)$$

$$v_t \bar{\omega} = y_t^C - c_t - i_t - g_t - n_t \psi_t \quad (44)$$

$$\rho_t^n = 1 - \Phi \left(\frac{\ln \bar{f}_t}{\sigma_w} \right) \quad (45)$$

$$\bar{f}_t = p_t^w (1 - \alpha) \frac{y_t^H}{n_t} - w_t h_t + \frac{\bar{\omega}}{e_t} \quad (46)$$

$$\psi_t = \frac{\exp \left(\frac{\sigma_w^2}{2} \right) \Phi \left(\frac{\ln \bar{f}_t - \sigma_w^2}{\sigma_w} \right)}{1 - \rho_t^n} \quad (47)$$

$$y_t^m = c_t + i_t + g_t + x_t^{H*} + y_t^{Co} - y_t \quad (48)$$

Taylor rule (1 equation):

$$\frac{R_t}{R} = \left(\frac{R_{t-1}}{R} \right)^{\rho_R} \left[\left(\frac{\pi_t}{\bar{\pi}} \right)^{\alpha_\pi} \left(\frac{y_t}{y_{t-1}} \right)^{\alpha_y} \right]^{1-\rho_R} \exp(\varepsilon_t^R) \quad (49)$$

Rest of the world (4 equations):

$$\xi_t = \bar{\xi} \exp \left[-\psi \frac{rer_t b_t^* - rer \times b^*}{rer \times b^*} + \frac{\zeta_t - \zeta}{\zeta} \right] \quad (50)$$

$$\frac{rer_t}{rer_{t-1}} = \frac{\pi_t^s \pi_t^*}{\pi_t} \quad (51)$$

$$\lambda_t = \frac{\beta}{a_t^\sigma} R_t^* \xi_t E_t \left\{ \frac{\varrho_{t+1}}{\varrho_t} \frac{\pi_{t+1}^S \lambda_{t+1}}{\pi_{t+1}} \right\} \quad (52)$$

$$x_t^{H*} = o^* \left(\frac{p_t^H}{rer_t} \right)^{-\eta^*} y_t^* \quad (53)$$

Aggregation and market clearing (7 equations):

$$y_t^H = x_t^H + x_t^{H*} \quad (54)$$

$$y_t^F = x_t^F \quad (55)$$

$$\Delta_t^H = (1 - \theta_H) (\tilde{p}_t^H)^{-\epsilon_H} + \theta_H \left(\frac{p_{t-1}^H}{p_t^H} \frac{\Gamma_t^H}{\pi_t} \right)^{-\epsilon_H} \Delta_{t-1}^H \quad (56)$$

$$\Delta_t^F = (1 - \theta_F) (\tilde{p}_t^F)^{-\epsilon_F} + \theta_F \left(\frac{p_{t-1}^F}{p_t^F} \frac{\Gamma_t^F}{\pi_t} \right)^{-\epsilon_F} \Delta_{t-1}^F \quad (57)$$

$$b_t^* + (1 - \chi) p_t^{Co*} y_t^{Co} = \frac{b_{t-1}^*}{a_{t-1}} \frac{R_{t-1}^* \xi_t}{\pi_t^*} + \frac{tb_t}{rer_t} \quad (58)$$

$$tb_t = p_t^H x_t^{H*} + rer_t p_t^{Co*} y_t^{Co} - rer_t y_t^m \quad (59)$$

$$p_t^Y y_t = c_t + i_t + g_t + tb_t \quad (60)$$

B Steady State of the Search Model

We show how to compute the steady state of the model for given values $h, u, \rho = p^{EU} / (1 - p^{UE})$, $s_{\rho x} = \rho / \rho_x$, e , p^H , $s^{tb} = tb / (p^Y y)$, $s^g = g / (p^Y y)$ and $s^{Co} = rer \times p^{Co*} y^{Co} / (p^Y y)$. The parameters $\bar{m}$, $\bar{\rho}_x$, $\bar{\omega}$, $\bar{g}$, $\bar{y}^{Co}$, $\bar{\kappa}$, $\bar{\pi}^*$, o^* and $\bar{b}$ are determined endogenously while the values of the remaining parameters are taken as given.

From the exogenous processes $x_t = \{\varrho_t, \varpi_t, z_t, a_t, R_t^*, \zeta_t, y_t^*, p_t^{Co*}\}$,

$$x = \bar{x}.$$

From (6) and (7),

$$\Theta = 1.$$

From (50),

$$\xi = \bar{\xi}.$$

From (49),

$$\pi = \bar{\pi}.$$

From (4),

$$R = \frac{\pi a^\sigma}{\beta}.$$

From (2),

$$q = \varpi^{-1}.$$

From (3),

$$r^K = q \left(\frac{a^\sigma}{\beta} - 1 + \delta \right).$$

From (52),

$$\pi^S = \frac{a^\sigma \pi}{\beta R^* \xi}.$$

From (51),

$$\pi^* = \bar{\pi}^* = \frac{\pi}{\pi^s}.$$

From (40) and (41),

$$\tilde{p}_t^H = 1, \quad \tilde{p}_t^F = 1.$$

From (56) and (57),

$$\Delta^H = (\tilde{p}^H)^{-\epsilon_H}, \quad \Delta^F = (\tilde{p}^F)^{-\epsilon_F}.$$

From (36)-(37) and (38)-(39),

$$mc^H = \left(\frac{\epsilon_H - 1}{\epsilon_H} \right) \tilde{p}^H, \quad mc^F = \left(\frac{\epsilon_F - 1}{\epsilon_F} \right) \tilde{p}^F.$$

From (34),

$$p^w = p^H mc^H.$$

From (10),

$$n = 1 - u.$$

From $s_{\rho x} = \rho^x / \rho$,

$$\rho^x = \bar{\rho}^x = s_{\rho x} \rho.$$

From (13),

$$\rho^n = \frac{\rho - \rho^x}{1 - \rho^x}.$$

From (45),

$$\bar{f} = \exp \left(\sigma_w \Phi^{-1} (1 - \rho^n) \right).$$

From (47),

$$\psi = \frac{\exp \left(\frac{\sigma_w^2}{2} \right) \Phi \left(\frac{\ln \bar{f} - \sigma_w^2}{\sigma_w} \right)}{1 - \rho^n}.$$

From (42) and (46),

$$\bar{\omega} = ea^{1-\sigma}\beta(1-\rho)(\bar{f}-\psi).$$

From (9) and (12),

$$v = \frac{\rho n}{e(1-\rho)}.$$

From (12),

$$m = ev.$$

From (8),

$$\bar{m} = mv^{\mu-1}u^{-\mu}.$$

From (11),

$$s = m/u.$$

From (32) and (43),

$$k = a^2hn\left(\frac{\alpha p^w z}{\Delta^H r^K}\right)^{\frac{1}{1-\alpha}}.$$

From (32),

$$y^H = \frac{z\left(\frac{k}{a}\right)^\alpha (anh)^{1-\alpha}}{\Delta^H}.$$

From (46),

$$w = \frac{1}{h}\left(p^w(1-\alpha)\frac{y^H}{n} + \frac{\bar{\omega}}{e} - \bar{f}\right).$$

From (25)-(27),

$$w^* = w\left(\frac{1-(1-\rho)(1-s)a^{\alpha_w-1}\theta^w}{1-\theta^w}\right)\left(1+\frac{(1-\rho)s}{n}\frac{a^{\alpha_w-1}\theta^w}{1-a^{\alpha_w-1}\theta^w}\right)^{-1}.$$

From (27),

$$\tilde{w} = \frac{1-\theta^w}{1-a^{\alpha_w-1}\theta^w}w^*.$$

From (25),

$$\bar{w} = \frac{w}{\theta^w} - \frac{1-\theta^w}{\theta^w}w^*.$$

From (19),

$$\epsilon = \frac{h}{1-\theta^w a^{\alpha_w-\sigma}\beta(1-\rho)}.$$

From (20),

$$\mu = \frac{h}{1-\theta^w a^{\alpha_w-\sigma}\beta(1-\rho)(1-e)}.$$

From (21),

$$\mathcal{F}^* = p^w (1 - \alpha) \frac{y^H}{n} - w^* h - \psi + \frac{\bar{\omega}}{e}.$$

From (22),

$$\mathcal{W}^* = \frac{\varphi \epsilon}{(1 - \varphi) \mu} \mathcal{F}^*.$$

From (17),

$$\Delta^1 = \left(\frac{a^{\alpha_w - 1} \pi w / w^* - 1}{1 - a^{1 - \sigma} \theta^w \beta} \right) w^* h.$$

From (18),

$$\Delta^2 = \left(\frac{a^{\alpha_w - 1} \pi - 1}{1 - a^{1 - \sigma} \theta^w \beta} \right) w^* h.$$

From (14)-(16), (23) and (24),

$$\mathcal{U} = \frac{\alpha_7 - \alpha_3 + \alpha_5 + a^{1 - \sigma} \beta s (1 - \rho) \alpha_7}{\alpha_8 + \alpha_4 - \alpha_6 + a^{1 - \sigma} \beta s (1 - \rho) \alpha_8 - a^{1 - \sigma} \beta}$$

where

$$\alpha_1 = \frac{w^* h - \frac{\Theta \kappa h^{1 + \phi}}{\lambda(1 + \phi)} + a^{1 - \sigma} \beta (1 - \rho) \theta^w \Delta^2}{1 - a^{1 - \sigma} \beta (1 - \rho)}, \quad \alpha_2 = \frac{a^{1 - \sigma} \beta \rho}{1 - a^{1 - \sigma} \beta (1 - \rho)},$$

$$\alpha_3 = wh - \frac{\Theta \kappa h^{1 + \phi}}{\lambda(1 + \phi)} + a^{1 - \sigma} \beta (1 - \rho) (\theta^w \Delta^1 + \alpha_1),$$

$$\alpha_4 = ((1 - \rho) \alpha_2 + \rho) a^{1 - \sigma} \beta, \quad \alpha_5 = -a^{1 - \sigma} \beta s (1 - \rho) \alpha_3, \quad \alpha_6 = 1 - a^{1 - \sigma} \beta (1 + s (1 - \rho) (\alpha_4 - 1)),$$

$$\alpha_7 = \frac{\left(1 + a^{1 - \sigma} \beta (1 - \rho) \theta^w \frac{\varphi}{1 - \varphi} \frac{\epsilon}{\mu} (1 - \Gamma^W) \right) w^* h - \frac{\Theta \kappa h^{1 + \phi}}{\lambda(1 + \phi)} - (1 - a^{1 - \sigma} \beta (1 - \rho)) \mathcal{W}^* - \alpha_5}{a^{1 - \sigma} \beta s (1 - \rho)}, \quad \alpha_8 = \frac{\alpha_6}{a^{1 - \sigma} \beta s (1 - \rho)}$$

From (15),

$$\mathcal{E}^* = \alpha_1 + \alpha_2 \mathcal{U}$$

From (14),

$$\bar{\mathcal{E}} = \alpha_3 + \alpha_4 \mathcal{U}$$

From (16),

$$\bar{b} = \alpha_5 + \alpha_6 \mathcal{U}$$

From (24),

$$\bar{\mathcal{W}} = \alpha_7 - \alpha_8 \mathcal{U}$$

From (36) and $\Xi = \beta / a^\sigma$,

$$f^H = \frac{(\tilde{p}^H)^{-\epsilon_H} m c^H y^H}{1 - \beta a^{1 - \sigma} \theta_H}.$$

From (5),

$$i = \frac{1 - (1 - \delta)/a}{\varpi} k.$$

From (29)-(31),

$$p^F = \left(\frac{1}{o} - \frac{1-o}{o} \right) (p^H)^{1-\eta} \Big)^{1/(1-\eta)}.$$

From (35),

$$rer = p^F mc^F.$$

From nominal GDP equal to $p^Y y = p^H y^H + p^F y^F + rer \times p^{Co*} y^{Co} - rer \times y^m - \psi n - \bar{\omega} v$,

$$p^Y y = \frac{p^H y^H + (\psi n + \bar{\omega} v) \left(p^F (1 - mc^F \Delta^F) o (p^F)^{-\eta} - 1 \right)}{1 - s^{Co} - p^F (1 - mc^F \Delta^F) o (p^F)^{-\eta} (1 - s^{tb})}.$$

From $s^{tb} = tb/(p^Y y)$, $s^g = g/(p^Y y)$ and $s^{Co} = rer p^{Co*} y^{Co}/(p^Y y)$,

$$tb = s^{tb} p^Y y, \quad g = \bar{g} = s^g p^Y y, \quad y^{Co} = \bar{y}^{Co} = \frac{s^{Co} p^Y y}{rer \times p^{Co*}}.$$

From (30), (44), (54) and (60),

$$x^{H*} = y^H - (1 - o) (p_t^H)^{-\eta} (c + i + g + n\psi + v\bar{\omega})$$

From (59), (33) and (55),

$$x_t^F = \frac{p^H x^{H*} + rer p^{Co*} y^{Co} - tb}{rer \times \Delta_t^F}.$$

From (31),

$$y^C = \frac{x^F}{o} (p^F)^\eta.$$

From (44),

$$c = y^C - i - g - n\psi - v\bar{\omega}.$$

From (1),

$$\lambda = \left(c - \zeta \frac{c}{a} \right)^{-\sigma}.$$

From (28),

$$\kappa = \bar{\kappa} = \frac{p^w \lambda (1 - \alpha)^2 y^H}{\Theta h^{1+\phi} n}.$$

From (7),

$$\tilde{\chi} = \left(c - \zeta \frac{c}{a} \right)^\sigma.$$

From (53),

$$o^* = \frac{x^{H*}}{y^*} \left(\frac{p^H}{rer} \right)^{\eta^*}.$$

From (58),

$$b^* = \frac{tb - (1 - \chi) rer \times p^{Co*} y^{Co}}{rer (1 - R^* \xi / a \pi^*)}.$$

From (54),

$$x^H = y^H - x^{H*}.$$

From (55),

$$y^F = x^F.$$

From (38),

$$f^F = \frac{(\tilde{p}^F)^{-\epsilon_F} y^F m c^F}{1 - \beta a^{1-\sigma} \theta_F}.$$

From (33),

$$y^m = y^F \Delta^F.$$

From (48),

$$y = c + i + g + x^{H*} + y^{Co} - y^m.$$

From (60),

$$p^Y = \frac{c + i + g + tb}{y}.$$

C Equilibrium Conditions of the Calvo Wages Model

The rational expectations equilibrium of the stationary version of the model is the set of sequences

$$\{\lambda_t, c_t, \Theta_t, \tilde{\chi}_t, h_t, h_t^d, w_t, \tilde{w}_t, mc_t^W, f_t^W, \Delta_t^W, \gamma_t^W, i_t, k_t, r_t^K, q_t, y_t, y_t^C, y_t^F, y_t^H, x_t^F, x_t^H, x_t^{H*}, R_t, \xi_t, \pi_t, \pi_t^S, rer_t, p_t^H, \tilde{p}_t^H, p_t^F, \tilde{p}_t^F, p_t^Y, p_t^m, mc_t^H, f_t^H, \Delta_t^H, mc_t^F, f_t^F, \Delta_t^F, b_t^*, imp_t, tb_t\}_{t=0}^\infty,$$

(43 variables) such that for given initial values and exogenous sequences

$$\{\kappa_t, \varrho_t, \varpi_t, z_t, a_t, \zeta_t^o, \zeta_t^u, R_t^*, \pi_t^*, p_t^{Co*}, y_t^{Co}, y_t^*, g_t\}_{t=0}^\infty,$$

conditions (1)-(7), (29)-(31), (33)-(41), (43), (48)-(60), and the following conditions are satisfied:

$$w_t mc_t^W = \Theta_t \kappa_t \frac{h_t^\phi}{\lambda_t}, \tag{61}$$

$$f_t^W = mc_t^W \tilde{w}_t^{-\epsilon_W} h_t^d + \frac{\beta}{a_t^{\sigma-1}} \theta_W E_t \left\{ \frac{\varrho_{t+1}}{\varrho_t} \frac{\lambda_{t+1}}{\lambda_t} \left(\frac{\gamma_t^W}{a_t \pi_{t+1}} \right)^{-\epsilon_W} \left(\frac{\tilde{w}_t}{\tilde{w}_{t+1}} \right)^{-\epsilon_W} \left(\frac{w_t}{w_{t+1}} \right)^{-1-\epsilon_W} f_{t+1}^W \right\}, \quad (62)$$

$$f_t^W = \tilde{w}_t^{1-\epsilon_W} h_t^d \left(\frac{\epsilon_W - 1}{\epsilon_W} \right) + \frac{\beta}{a_t^{\sigma-1}} \theta_W E_t \left\{ \frac{\varrho_{t+1}}{\varrho_t} \frac{\lambda_{t+1}}{\lambda_t} \left(\frac{\gamma_t^W}{a_t \pi_{t+1}} \right)^{1-\epsilon_W} \left(\frac{\tilde{w}_t}{\tilde{w}_{t+1}} \right)^{1-\epsilon_W} \left(\frac{w_t}{w_{t+1}} \right)^{-\epsilon_W} f_{t+1}^W \right\}, \quad (63)$$

$$1 = (1 - \theta_W) \tilde{w}_t^{1-\epsilon_W} + \theta_W \left(\frac{w_{t-1}}{w_t} \frac{\gamma_t^W}{a_{t-1} \pi_t} \right)^{1-\epsilon_W}, \quad (64)$$

$$\Delta_t^W = (1 - \theta_W) \tilde{w}_t^{-\epsilon_W} + \theta_W \left(\frac{w_{t-1}}{w_t} \frac{\gamma_t^W}{a_{t-1} \pi_t} \right)^{-\epsilon_W} \Delta_{t-1}^W, \quad (65)$$

$$y_t^H \Delta_t^H = z_t \left(\frac{k_{t-1}}{a_{t-1}} \right)^\alpha (a_t h_t^d)^{1-\alpha}, \quad (66)$$

$$w_t = p_t^m \alpha \frac{y_t^H}{h_t^d} a_{t-1}, \quad (67)$$

$$h_t = h_t^d \Delta_t^W, \quad (68)$$

$$y_t^C = c_t + i_t + g_t, \quad (69)$$

$$\gamma_t^W = a^{\alpha_W} \pi_t^{\vartheta_W} \bar{\pi}^{1-\vartheta_W}. \quad (70)$$

D Steady State of the Calvo Wages Model

We solve for the steady state for given values of h , p^H , $s^{tb} = tb/(p^Y y)$, $s^g = g/(p^Y y)$ and $s^{Co} = rer \times p^{Co*} y^{Co}/(p^Y y)$. The parameters $\bar{\pi}^*$, $\bar{\kappa}$, o^* , $\bar{g}$ and $\bar{y}^{Co}$ are determined endogenously while the values of the remaining parameters are taken as given. The following equations are added to the steady state of the model with search frictions.

From (64),

$$\tilde{w} = \left(\frac{1 - \theta_W (\gamma^W / a \pi)^{1-\epsilon_W}}{1 - \theta_W} \right)^{\frac{1}{1-\epsilon_W}}.$$

From (65),

$$\Delta^W = \frac{1 - \theta_W}{1 - \theta_W (\gamma^W / a \pi)^{-\epsilon_W}} \tilde{w}^{-\epsilon_W}.$$

From (62)-(63),

$$mc^W = \left(\frac{\epsilon_W - 1}{\epsilon_W} \frac{1 - \beta a^{1-\sigma} (\gamma^W / a \pi)^{-\epsilon_W} \theta_W}{1 - \beta a^{1-\sigma} (\gamma^W / a \pi)^{1-\epsilon_W} \theta_W} \right) \tilde{w}.$$

From (68),

$$h^d = h/\Delta^W.$$

From (62),

$$f^W = \tilde{w}^{-\epsilon_W} h^d m c^W / \left(1 - \beta a^{1-\sigma} (\gamma^W / a\pi)^{-\epsilon_W} \theta_W\right).$$

From (56), (2) and (67),

$$w = \left[\frac{\alpha^\alpha (1-\alpha)^{1-\alpha} p^H m c^H z a^{1-\alpha}}{(r^K)^\alpha} \right]^{\frac{1}{1-\alpha}}.$$

From (56) and (67),

$$k = \frac{\alpha a w h^d}{(1-\alpha) r^K}.$$

From (66),

$$y^H = z (k/a)^\alpha (a h^d)^{1-\alpha} / \Delta^H.$$

From (69),

$$c = y^C - g - i.$$

From GDP equal to value added,

$$p^Y y = p^H y^H + p^Y y s^{Co} + p^F (1 - m c^F \Delta^F) o (p^F)^{-\eta} (1 - s^{tb}) p^Y y.$$

From (61),

$$\kappa = m c^W \lambda w / (\Theta h^\phi).$$

The remaining equations are the same as in the model with search frictions, except for the equations corresponding to the labor market variables from the search model which are eliminated.

E Detailed Results and Robustness Checks

Table 1: Calibrated Parameters and Targeted Steady State Values.

Parameter	Description	Value	Source
<i>Search model</i>			
u	Unemployment rate in st. st.	0.08	Average (1987-2014)
e	Firm matching rate	0.7	Den Haan et al. (2000)
ρ	Total separation rate	0.0755	Jones and Naudon (2009)
ρ^x	Exog. separation rate	$\frac{2}{3}\rho$	Den Haan et al. (2000)
$\mu_{\tilde{c}}$	Log-normal mean of $\tilde{c}$	0	Normalization
<i>Standard model</i>			
ϵ_W	E. o. s. wages	11	Medina and Soto (2007)
<i>Common parameters</i>			
σ	Inverse intertemporal e. o. s.	1	Medina and Soto (2007)
α	Capital share in production	1-0.66	Medina and Soto (2007)
δ	Capital depreciation	0.06/4	Medina and Soto (2007)
ϵ_H	E. o. s. domestic aggregate	11	Medina and Soto (2007)
ϵ_F	E. o. s. imported aggregate	11	Medina and Soto (2007)
α_W	Indexation parameter	1	Medina and Soto (2007)
o	Share of F in Y^C	0.32	Average (1987-2014)
χ	Gov. share in commodity sector	0.61	Average (1987-2014)
s^{tb}	Trade balance to GDP in st. st.	0.04	Average (1987-2014)
s^g	Gov. cons. to GDP in st. st.	0.11	Average (1987-2014)
s^{Co}	Commod. prod. to GDP in st. st.	0.10	Average (1987-2014)
$\bar{\pi}$	Inflation in st. st.	3% p.a.	Inflation target in Chile
p^H	Relative price of H in st. st.	1	Normalization
h	Hours per worker in st. st.	0.3	Normalization
$\bar{a}$	Long-run growth	2% p.a.	Albagli et al. (2015)
β	Subjective discount factor	0.9995	MPR approx. 5%
R^*	Foreign rate in st. st.	4.5% p.a.	Fuentes and Gredig (2008)
ξ	Country premium in st. st.	1.5% p.a.	Average (1987-2014)

Note: All rates are annualized figures.

Table 2: Marginal Data Densities.

	Search				Standard
	With ρ_t^x	W/out ρ_t^x	Low ρ_t^n	W/out ρ_t^x , low ρ_t^n	
$\log p(X^T \text{ without } u^T \theta)$	-1453.17	-1456.04	-1443.33	-1443.18	-1473.38
$\log p(X^T \theta)$	-1505.99	-1525.49	-1488.75	-	-1473.38

Note: X^T denotes the full data set, X^T without u^T the set excluding the unemployment rate. For the model with search frictions, we also compute the marginal likelihoods shutting down the exogenous separation shock (ρ_t^x), and lowering endogenous separations in equilibrium. The marginal data densities are Laplace approximations at the mean of the posterior distribution.

Table 3: Second Moments.

Variable	Description	s.d. (%)			AC order 1			Correl. Non-mining GDP		
		Data	Search	Standard	Data	Search	Standard	Data	Search	Standard
$\Delta \log Y^{NM}$	Non-mining GDP	1.14	1.66	1.92	0.20	0.00	0.61	1.00	1.00	1.00
$\Delta \log C$	Consumption	1.21	1.28	1.72	0.36	0.47	0.66	0.64	0.21	0.56
$\Delta \log I$	Investment	3.75	4.69	6.02	0.20	0.47	0.74	0.22	0.60	0.75
π	Inflation	0.69	0.79	1.23	0.60	0.49	0.63	-0.33	-0.31	-0.20
R	MPR	0.42	0.47	0.85	0.88	0.88	0.94	-0.35	-0.07	-0.22
rer	Real exch. rate	5.17	6.69	10.29	0.75	0.76	0.89	-0.17	-0.38	-0.18
ξ	EMBIG Chile	0.15	0.26	0.27	0.83	0.94	0.95	-0.24	-0.10	-0.13
$\Delta \log W$	Real wage	0.58	0.69	1.25	0.36	0.44	0.48	0.24	0.48	0.59
$h \times n$	Total hours worked	1.87	1.97	7.61	0.73	0.90	0.89	0.06	0.13	0.1
u	Unemployment rate	1.43	1.37	—	0.96	0.87	—	0.11	-0.14	—

Note: The model moments are the theoretical moments at the posterior mean.

Table 4: Estimated Parameters.

Param.	Description	Posterior						
		Prior			Search		Standard	
		Dist.	Mean	s.d.	Mean	90% HPDI	Mean	90% HPDI
v	Wealth effect size	beta	0.5	0.25	0.10	[0.00, 0.21]	0.34	[0.00, 0.70]
ϕ	Inv. elast. of h	norm	2	2	5.09	[2.90, 7.30]	1.00	[0.00, 2.16]
ς	Habit formation	beta	0.7	0.1	0.75	[0.66, 0.83]	0.82	[0.74, 0.89]
ψ	Country prem. elast.	invg	0.01	Inf	0.005	[0.003, 0.007]	0.005	[0.003, 0.006]
η	E. o. s. H and F	invg	1.5	0.25	3.53	[2.87, 4.18]	1.74	[1.36, 2.09]
η^*	RER elast. of X^{H*}	invg	0.25	0.1	0.23	[0.13, 0.32]	0.18	[0.12, 0.24]
γ	Inv. adj. cost	norm	4	1.5	0.81	[0.11, 1.77]	5.48	[3.59, 7.44]
φ	Bargaining power	beta	0.5	0.1	0.27	[0.09, 0.45]	—	—
σ_w	s.d. of F	norm	0.1	0.05	0.30	[0.25, 0.36]	—	—
μ	Match elast.	beta	0.5	0.1	0.75	[0.65, 0.84]	—	—
θ_W	Nom. W stickiness	beta	0.5	0.1	0.78	[0.67, 0.89]	0.90	[0.87, 0.95]
ϑ_W	Index. past infl. W	beta	0.5	0.15	0.24	[0.10, 0.36]	0.40	[0.21, 0.60]
θ_H	Calvo prob. H	beta	0.5	0.1	0.33	[0.24, 0.42]	0.56	[0.50, 0.61]
ϑ_H	Index. past infl. H	beta	0.5	0.15	0.51	[0.27, 0.76]	0.21	[0.08, 0.35]
θ_F	Calvo prob. F	beta	0.5	0.1	0.82	[0.77, 0.86]	0.57	[0.50, 0.63]
ϑ_F	Index. past infl. F	beta	0.5	0.15	0.64	[0.48, 0.80]	0.54	[0.29, 0.77]
ρ_R	MPR rule R_{t-1}	beta	0.75	0.1	0.83	[0.79, 0.87]	0.85	[0.82, 0.89]
α_π	MPR rule π_t	norm	1.5	0.1	1.55	[1.40, 1.70]	1.50	[1.36, 1.64]
α_y	MPR rule Δy_t	norm	0.125	0.05	0.15	[0.08, 0.23]	0.13	[0.05, 0.21]
ρ_ϱ	AC cons. pref. sh.	beta	0.75	0.1	0.64	[0.50, 0.79]	0.74	[0.60, 0.88]
ρ_κ	AC labor pref. sh.	beta	0.75	0.1	0.77	[0.64, 0.91]	0.74	[0.58, 0.90]
ρ_{ρ^x}	AC exo. sep. sh.	beta	0.75	0.1	0.88	[0.81, 0.95]	—	—
ρ_ϖ	AC inv. sh.	beta	0.75	0.1	0.65	[0.51, 0.79]	0.85	[0.78, 0.92]
ρ_z	AC temp. TFP sh.	beta	0.75	0.1	0.72	[0.59, 0.85]	0.50	[0.39, 0.61]
ρ_a	AC perm. TFP sh.	beta	0.375	0.1	0.35	[0.21, 0.49]	0.32	[0.20, 0.43]
ρ_{ζ^o}	AC obs. risk sh.	beta	0.75	0.1	0.87	[0.81, 0.94]	0.85	[0.77, 0.93]
ρ_{ζ^u}	AC unobs. risk sh.	beta	0.75	0.1	0.77	[0.69, 0.86]	0.74	[0.64, 0.85]
$\rho_{y^{Co}}$	AC commodity sh.	beta	0.75	0.1	0.95	[0.91, 0.98]	0.94	[0.90, 0.98]
σ_ϱ	s.d. cons. pref. sh.	invg	0.01	Inf	0.04	[0.03, 0.05]	0.06	[0.04, 0.08]
σ_κ	s.d. labor pref. sh.	invg	0.01	Inf	0.03	[0.02, 0.04]	0.13	[0.03, 0.23]
σ_{ρ^x}	s.d. separation sh.	invg	0.01	Inf	0.15	[0.10, 0.20]	—	—
σ_ϖ	s.d. inv. shock	invg	0.01	Inf	0.03	[0.01, 0.05]	0.08	[0.05, 0.10]
σ_z	s.d. temp. TFP sh.	invg	0.01	Inf	0.007	[0.005, 0.009]	0.019	[0.015, 0.022]
σ_a	s.d. perm. TFP sh.	invg	0.01	Inf	0.003	[0.002, 0.004]	0.019	[0.015, 0.023]
σ_{ζ^o}	s.d. obs. risk sh.	invg	0.003	Inf	0.0008	[0.0006, 0.0009]	0.001	[0.001, 0.001]
σ_{ζ^u}	s.d. unobs. risk sh.	invg	0.003	Inf	0.007	[0.004, 0.01]	0.009	[0.005, 0.013]
σ_R	s.d. MPR shock	invg	0.003	Inf	0.0017	[0.0014, 0.0019]	0.0016	[0.0013, 0.0019]
$\sigma_{y^{Co}}$	s.d. commodity sh.	invg	0.01	Inf	0.07	[0.04, 0.09]	0.07	[0.05, 0.09]

Note: The results are based on 500,000 draws from the posterior distribution using the Metropolis-Hastings (MH) algorithm, dropping the first 250,000 draws to achieve convergence. The average acceptance rate of the MH algorithm was approximately 25% for each model. HPDI are the highest posterior density intervals. The priors for the parameters ϕ and α_π were truncated at 0 and 1, respectively. The computations were conducted using Dynare 4.5.5.

Table 5: Variance Decomposition.

Shock																			
Variable	TFP trans. z	TFP perm. a	Inv. tech. ϖ	Total tech. sh.	Cons. pref. ϱ	Lab. sup. κ	Total pref. sh.	MP rate e^R	Gov. cons. g	Risk obs. ζ^o	Risk unobs. ζ^u	Total risk sh.	For. rate R^*	For. infl. π^*	Co. price p^{Co*}	For. dem. y^*	Total for. sh.	Co. prod. y^{Co}	Exo. sep. ρ^x
A. Search																			
y^{NM}	19	5	10	34	5	5	10	0	0	0	7	7	20	6	11	0	37	2	9
c	1	3	2	5	16	0	16	0	0	0	8	8	9	13	41	0	64	6	0
i	2	0	19	22	5	1	6	1	0	1	18	19	31	8	10	0	49	2	1
π	24	2	33	60	3	4	7	12	0	0	7	7	3	4	1	0	9	0	4
R	6	1	40	48	4	1	5	7	0	0	10	11	12	6	9	0	27	1	1
rer	1	0	3	5	0	0	1	1	0	2	45	46	19	20	6	0	46	1	0
ξ	0	0	2	2	0	0	0	0	0	38	3	40	27	16	11	0	54	2	0
w	5	2	2	9	2	0	2	0	0	1	14	15	26	13	29	0	68	5	2
$h \times n$	3	1	16	19	5	22	27	2	0	0	4	5	7	4	10	0	22	2	24
u	3	1	17	21	1	1	2	3	0	0	3	3	9	2	5	0	16	1	53
B. Standard																			
y^{NM}	2	2	79	83	3	7	9	1	0	0	1	1	3	1	2	0	6	0	—
c	0	12	10	23	22	1	23	0	0	0	5	5	8	8	29	0	44	4	—
i	0	14	66	80	3	1	4	0	0	0	2	2	7	2	4	0	13	1	—
π	18	17	20	55	4	1	5	5	0	0	20	20	7	7	2	0	15	0	—
R	4	12	43	60	6	1	8	3	0	0	12	13	9	4	4	0	17	0	—
rer	1	7	21	30	1	2	3	1	0	0	19	19	14	12	19	0	45	3	—
ξ	0	4	11	15	1	0	1	0	0	28	5	33	11	16	21	0	47	4	—
w	1	23	7	32	2	4	6	0	0	0	7	7	11	9	31	0	51	4	—
$h \times n$	9	24	48	81	2	10	12	1	0	0	2	2	1	1	2	0	4	0	—

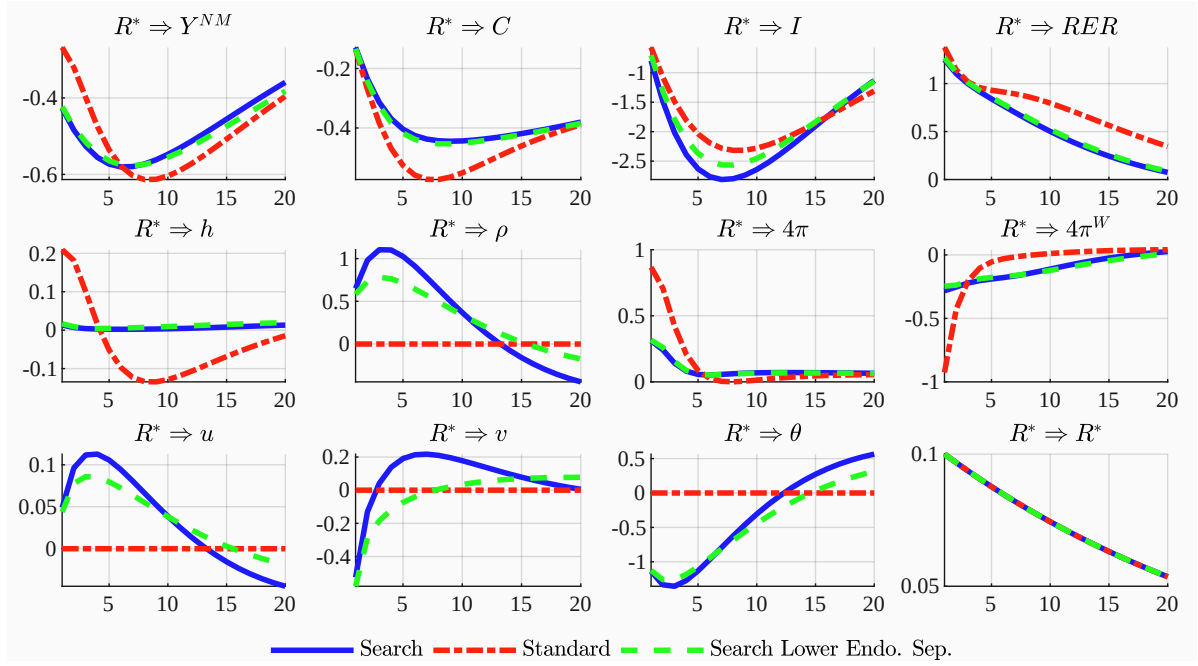
Note: The table entries are the fraction of the unconditional theoretical variances at the posterior mean (in %) explained by the shocks.

Table 6: Estimated Parameters: Standard Model with Smaller Capital Friction.

Param.	Description	Posterior						
		Prior			Search		Standard (Mean)	
		Dist.	Mean	s.d.	Mean	90% HPDI	Baseline	Fixed γ
v	Wealth effect size	beta	0.5	0.25	0.10	[0.00, 0.21]	0.34	0.0496
ϕ	Inv. elast. of h	norm	2	2	5.09	[2.90, 7.30]	1.00	2.0803
ς	Habit formation	beta	0.7	0.1	0.75	[0.66, 0.83]	0.82	0.8399
ψ	Country prem. elast.	invg	0.01	Inf	0.005	[0.003, 0.007]	0.005	0.0048
η	E. o. s. H and F	invg	1.5	0.25	3.53	[2.87, 4.18]	1.74	1.7573
η^*	RER elast. of X^{H*}	invg	0.25	0.1	0.23	[0.13, 0.32]	0.18	0.1794
γ	Inv. adj. cost	norm	4	1.5	0.81	[0.11, 1.77]	5.48	—
φ	Bargaining power	beta	0.5	0.1	0.27	[0.09, 0.45]	—	—
σ_w	s.d. of F	norm	0.1	0.05	0.30	[0.25, 0.36]	—	—
μ	Match elast.	beta	0.5	0.1	0.75	[0.65, 0.84]	—	—
θ_W	Nom. W stickiness	beta	0.5	0.1	0.78	[0.67, 0.89]	0.90	0.7922
ϑ_W	Index. past infl. W	beta	0.5	0.15	0.24	[0.10, 0.36]	0.40	0.5133
θ_H	Calvo prob. H	beta	0.5	0.1	0.33	[0.24, 0.42]	0.56	0.5464
ϑ_H	Index. past infl. H	beta	0.5	0.15	0.51	[0.27, 0.76]	0.21	0.2050
θ_F	Calvo prob. F	beta	0.5	0.1	0.82	[0.77, 0.86]	0.57	0.5849
ϑ_F	Index. past infl. F	beta	0.5	0.15	0.64	[0.48, 0.80]	0.54	0.5217
ρ_R	MPR rule R_{t-1}	beta	0.75	0.1	0.83	[0.79, 0.87]	0.85	0.8341
α_π	MPR rule π_t	norm	1.5	0.1	1.55	[1.40, 1.70]	1.50	1.4490
α_y	MPR rule Δy_t	norm	0.125	0.05	0.15	[0.08, 0.23]	0.13	0.1376
ρ_ϱ	AC cons. pref. sh.	beta	0.75	0.1	0.64	[0.50, 0.79]	0.74	0.7102
ρ_κ	AC labor pref. sh.	beta	0.75	0.1	0.77	[0.64, 0.91]	0.74	0.7862
ρ_{ρ^x}	AC exo. sep. sh.	beta	0.75	0.1	0.88	[0.81, 0.95]	—	—
ρ_{ϖ}	AC inv. sh.	beta	0.75	0.1	0.65	[0.51, 0.79]	0.85	0.9133
ρ_z	AC temp. TFP sh.	beta	0.75	0.1	0.72	[0.59, 0.85]	0.50	0.4641
ρ_a	AC perm. TFP sh.	beta	0.375	0.1	0.35	[0.21, 0.49]	0.32	0.2913
ρ_{ζ^o}	AC obs. risk sh.	beta	0.75	0.1	0.87	[0.81, 0.94]	0.85	0.8507
ρ_{ζ^u}	AC unobs. risk sh.	beta	0.75	0.1	0.77	[0.69, 0.86]	0.74	0.7490
$\rho_{y^{Co}}$	AC commodity sh.	beta	0.75	0.1	0.95	[0.91, 0.98]	0.94	0.9455
σ_ϱ	s.d. cons. pref. sh.	invg	0.01	Inf	0.04	[0.03, 0.05]	0.06	0.0689
σ_κ	s.d. labor pref. sh.	invg	0.01	Inf	0.03	[0.02, 0.04]	0.13	0.0636
σ_{ρ^x}	s.d. separation sh.	invg	0.01	Inf	0.15	[0.10, 0.20]	—	—
σ_{ϖ}	s.d. inv. shock	invg	0.01	Inf	0.03	[0.01, 0.05]	0.08	0.0284
σ_z	s.d. temp. TFP sh.	invg	0.01	Inf	0.007	[0.005, 0.009]	0.019	0.0182
σ_a	s.d. perm. TFP sh.	invg	0.01	Inf	0.003	[0.002, 0.004]	0.019	0.0178
σ_{ζ^o}	s.d. obs. risk sh.	invg	0.003	Inf	0.0008	[0.0006, 0.0009]	0.001	0.0007
σ_{ζ^u}	s.d. unobs. risk sh.	invg	0.003	Inf	0.007	[0.004, 0.01]	0.009	0.0087
σ_R	s.d. MPR shock	invg	0.003	Inf	0.0017	[0.0014, 0.0019]	0.0016	0.0017
$\sigma_{y^{Co}}$	s.d. commodity sh.	invg	0.01	Inf	0.07	[0.04, 0.09]	0.07	0.0670

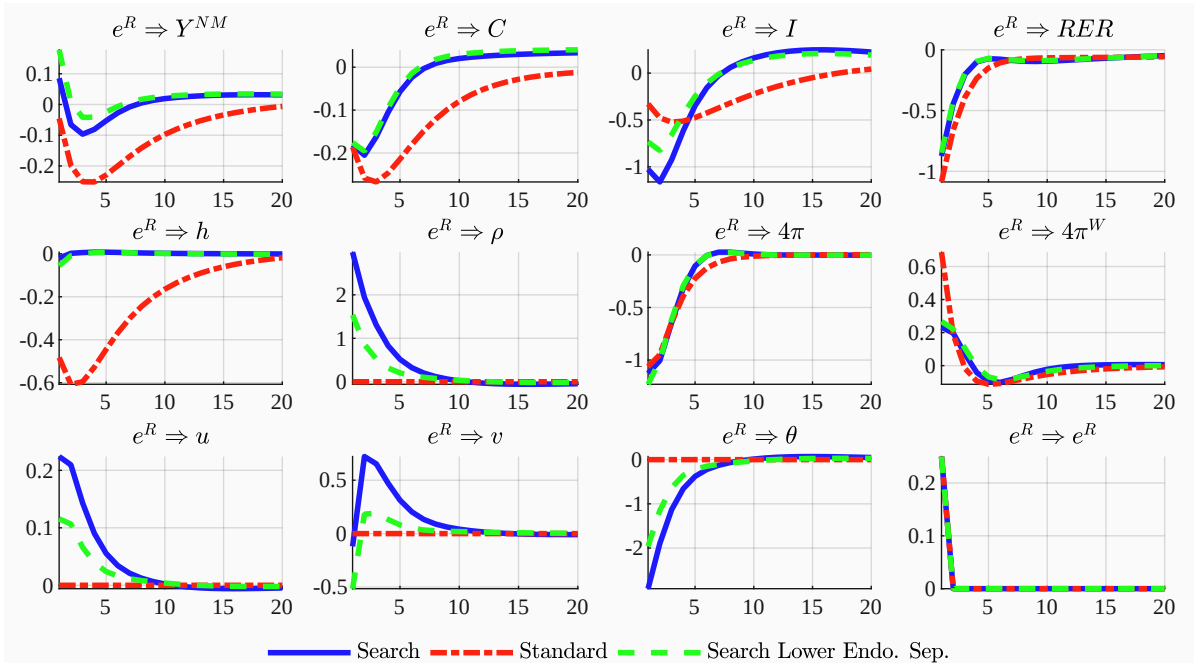
Note: The last two columns compare the posterior mean of two estimations of the standard model: the baseline estimation, and one in which we fix the elasticity of investment adjustment costs (γ) to the value estimated in the model with search frictions (0.81). See Table 4 for more details on the estimation of the model.

Figure 1: Impulse Responses to a Foreign Interest Rate Shock (R^*).



Note: The blue solid lines correspond to the model with search frictions, the red dash-dotted lines to the model with a standard labor market specification, and the green dashed lines to the model lower endogenous separations. In all cases the parameters associated to the shock process (persistence and volatility) are fixed at the prior mean, which is common across models. The variables are real non-mining GDP (Y^{NM}), household consumption (C), investment (I), the real exchange rate (rer), hours per worker (h), the separation rate (ρ), annualized CPI inflation (4π), annualized real wage growth ($4\pi^W$), the unemployment rate (u), vacancies (v), and labor market tightness (θ). All variables are expressed as percentage deviations from steady state.

Figure 2: Impulse Responses to a Domestic Monetary Policy Shock (e^R).



Note: See Figure 1.